

Scientific writing: Visualization, preparatory exercises

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Introduction. Please consider the following three exercises.

- In the first two, point out shortcomings of the current figure and propose improved visualization(s). You can either implement them explicitly with visualization software, or create sketches. The raw data is available with this problem sheet.
- In the third example, only the data is given along with some questions. Conceive figure(s) that provide answers to these questions.
- In all cases, also give a tentative caption that would go below the figure in a thesis or article.
- Please upload your solution as a single PDF via StudIP before day 3. The filename should be your name(s). This can be done in groups.

Exercise 1: algorithm runtimes.

The runtime of a distributed version of an algorithm (that performs some operation on images) is compared to a sequential single version. We want to learn how the runtimes between distributed and single versions compare, how the distributed runtime depends on the number of workers, and how the runtimes scale with image size. Figure 1 shows a preliminary version.

Exercise 2: scored and conceded goals in soccer.

We collected data on scored and conceded goals of some European soccer teams (in particular: their difference), in home and away matches. We want analyze if there is an indication of the home field advantage in the data. Do teams perform better when playing ‘at home’? Figure 2 shows a preliminary version.

Exercise 3: lengths in two species cultures.

In a fictive experiment, the lengths of plant leaves of two species is measured. The data is given in `lengths.csv` and contains a total of 500 measurements. The first column gives the time of the measurement in hours since the start of the experiment. The second column gives the length of the leaf of the specimen in millimeters. The third column indicates if the specimen is of type 0 or 1. Create figure(s) that show that the two species have different distributions of lengths, and that help to estimate the approximate growth rate of the leaves over time.

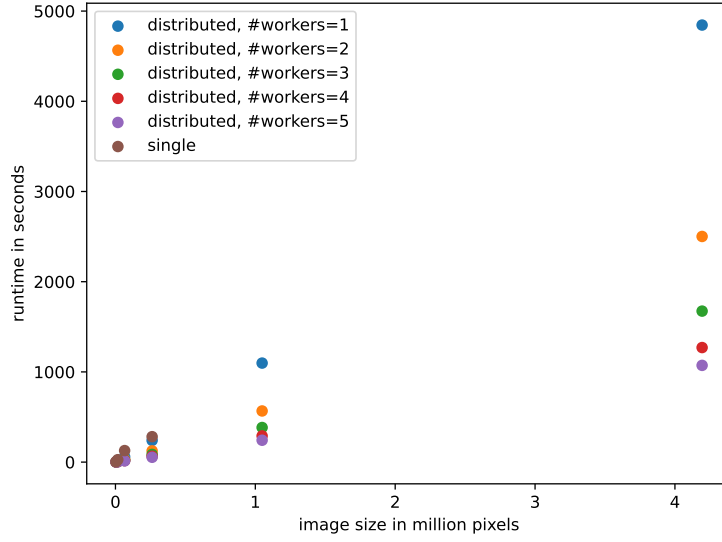


Figure 1: Runtimes of the algorithm, in the parallelized distributed version (with various numbers of workers) and the single sequential version, tested on images of varying size. As can be seen, the distributed version outperforms the single version and runtime decreases as the number of workers increases. Moreover, runtime increases with image size.

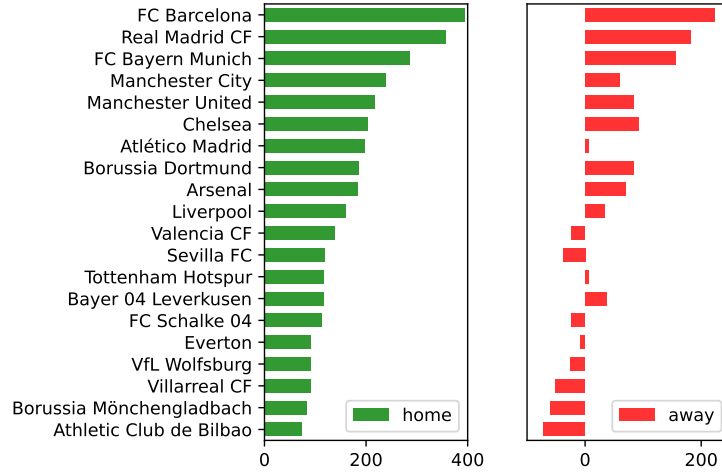


Figure 2: Difference between scored and conceded goals in home and away matches for several European clubs. As can be seen, the value is higher in home matches, indicating a clear home field advantage.